3.11 understand how the developing embryo is protected by amniotic fluid

3.12 understand the roles of oestrogen and testosterone in the development of secondary sexual characteristics.

b) Inheritance

Students will be assessed on their ability to:

3.13 understand that the nucleus of a cell contains chromosomes on which genes are located

3.14 understand that a gene is a section of a molecule of DNA and that a gene codes for a specific protein

3.15 describe a DNA molecule as two strands coiled to form a double helix, the strands being linked by a series of paired bases: adenine (A) with thymine (T), and cytosine (C) with guanine (G)

3.16 understand that genes exist in alternative forms called alleles which give rise to differences in inherited characteristics

3.17 understand the meaning of the terms: dominant, recessive, homozygous, heterozygous, phenotype, genotype and **codominance**

3.18 describe patterns of monohybrid inheritance using a genetic diagram

3.19 understand how to interpret family pedigrees

3.20 predict probabilities of outcomes from monohybrid crosses

3.21 understand that the sex of a person is controlled by one pair of chromosomes, XX in a female and XY in a male

3.22 describe the determination of the sex of offspring at fertilisation, using a genetic diagram

3.23 understand that division of a diploid cell by mitosis produces two cells which contain identical sets of chromosomes

3.24 understand that mitosis occurs during growth, repair, cloning and asexual reproduction

3.25 understand that division of a cell by meiosis produces four cells, each with half the number of chromosomes, and that this results in the formation of genetically different haploid gametes

3.26 understand that random fertilisation produces genetic variation of offspring

3.27 know that in human cells the diploid number of chromosomes is 46 and the haploid number is 23

3.28 understand that variation within a species can be genetic, environmental, or a combination of both

3.29 understand that mutation is a rare, random change in genetic material that can be inherited

3.30 describe the process of evolution by means of natural selection

3.31 understand that many mutations are harmful but some are neutral and a few are beneficial

3.32 understand that resistance to antibiotics can increase in bacterial populations, and appreciate how such an increase can lead to infections being difficult to control